



Artificial Intelligence: An introduction

Fabio Martínez, Ph.D - Gustavo Garzón, MSc.

Escuela de ingeniería de sistemas e informática
Biomedical Imaging, Vision and Learning Laboratory. BivL²ab
Machine learning Analysis and Computer Vision. M_LACV
Universidad Industrial de Santander

March 20, 2023

Universidad
Industrial de
Santander

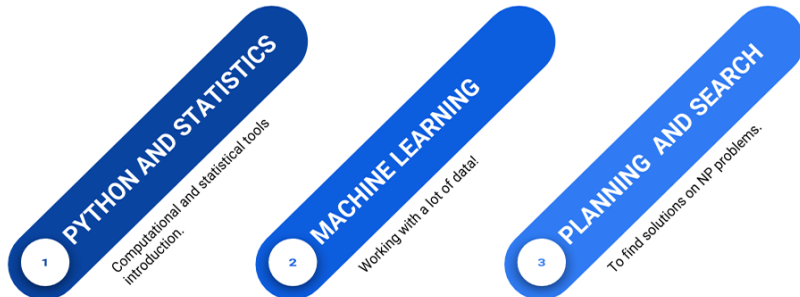


Biomedical Imaging, Vision and Learning Laboratory

Información del profesor

- ▶ Nombre: Fabio Martínez Carrillo
- ▶ Correo (contacto y comunicación):
famarcar@saber.uis.edu.co
- ▶ Nombre: Gustavo Garzón Villamizar
- ▶ Correo (contacto y comunicación):
gustavo.garzon@saber.uis.edu.co
- ▶ Contenido del curso:
<https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student>

Course outline



- ▶ **The course** is mainly based on the **coding of I.A concepts** and the **development of small applications**
- ▶ **The course** is focused on the analysis and study of **I.A algorithms**

I.A. Course



Course

It is about learning the fundamental tools to face problems and challenges of I.A. as well as some computational tools and theoretical concepts.

First module: Python and statistics introduction

- ▶ Introduction to Python
- ▶ Introduction to statistics
- ▶ Introduction to Visualization and data analytics



Second Module: Machine Learning

Classification and regression

- ▶ Naive Bayes
- ▶ Classification trees
- ▶ Support Vector Machines
- ▶ Introduction to Deep-learning

Non-supervised learning

- ▶ K-means
- ▶ DBScan



Third module: Planning and Searching

- ▶ Planning and searching
- ▶ Genetic algorithms (GA)
- ▶ Simulated annealing (SA)



Structure of the Course

- ▶ 30% Talleres (Problemsets)
- ▶ 10% Ejercicios clases. Las notas serán de acuerdo a los cuartiles de participación del curso. $Q_4=5$, $Q_3=4$, $Q_2=3$, $Q_1=2$. Para los que hayan estado por debajo de Q_1 se les promediara las notas de los ejercicios con respecto al total.



- ▶ 30% Parciales (Quizes)
- ▶ 30% Proyecto funcional IA.
 - ▶ 5 % primera entrega. Exploración de datos
 - ▶ 5 % segunda entrega. Validación de modelos clase
 - ▶ 20 % Entrega final. Incluye repositorio, presentación y avance del proyecto

PREVIOUS PROJECTS

- ▶ https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student/-/blob/master/projects/Proyectos_2018-2.md
- ▶ https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student/-/blob/master/projects/Proyectos_2019-1.md
- ▶ https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student/-/blob/master/projects/Proyectos_2019-2.md
https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student/-/blob/master/projects/Proyectos_2020-1.md

Some I.A. Applications: In classrooms

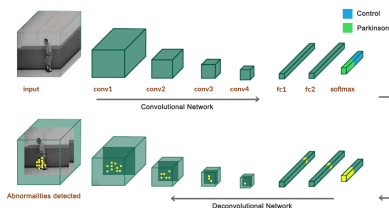


Video AI-classroom-CHINA

Some I.A. Applications: In Agriculture



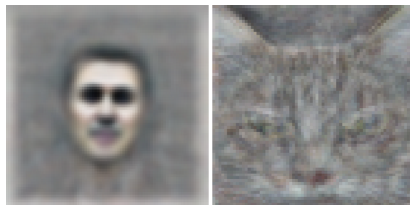
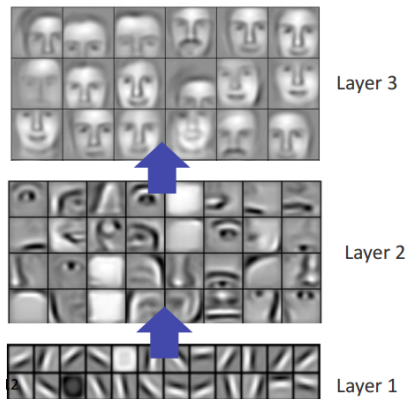
Salient Gait maps



Learning convolutional neuronal patterns

A 3D convolutional gait representation that produces salient spatio-temporal regions can potentially support and complement the diagnosis and the following of Parkinson's disease

Big challenges: How understand complex concepts?



Now, the machines can dream



A photograph of a Go match between a human player and a computer. The human player, wearing a dark suit, is seated on the left, looking at a board. The computer player, wearing a dark suit, is seated on the right, also looking at the board. A Go board is placed on a table between them. The background is dark. The text "Video 01" is overlaid in green in the upper right. The text "ALPHAGO" is overlaid in large white letters at the bottom.

Video 01

ALPHAGO

A photograph of a Go match between AlphaGo and Lee Sedol. Two players are seated at a table with a Go board. The player on the left is AlphaGo, and the player on the right is Lee Sedol. The Go board is visible on the table, and the players are focused on the game. The background is dark, and the lighting is focused on the players and the board. The text "Video 02" is overlaid in green on the right side of the image. The word "ALPHAGO" is written in large white letters at the bottom of the image.

Video 02

ALPHAGO

ImageNet Challenge

IMAGENET

IMAGENET Netflix 22:30, Bill Nye

- 1,000 object classes (categories).
- Images:
 - 1.2 M train
 - 100k test.

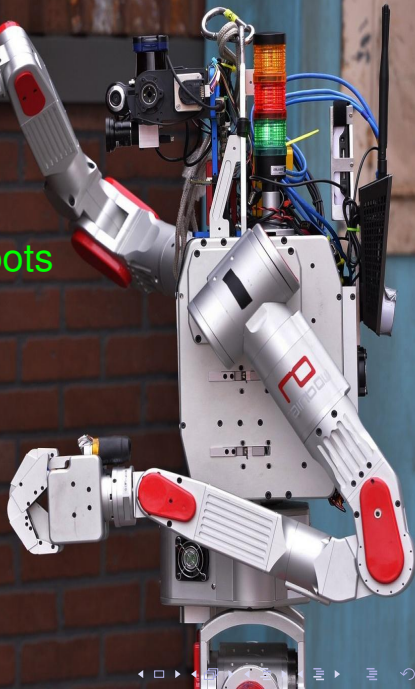


As support for decisions making

- ▶ Medicine
- ▶ Politics
- ▶ Laws



NOVA Rise of the Robots





y ahora ...Atlas. Google again!

Boston Dynamics

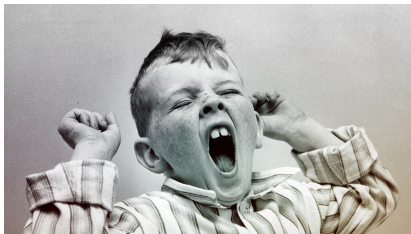
DARPA



Understanding Artificial Intelligence and Its Future

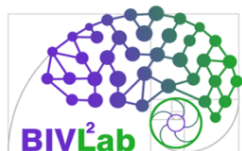


Thank you for your attention ...



... It's time to wake up

Universidad
Industrial de
Santander



Biomedical Imaging, Vision and Learning Laboratory